

Arman Javan Sekhavat Pishkhani

TEHRAN, IRAN

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EDUCATION

Master's Degree, Mechatronics Engineering

University of Tehran

2024–2027

TEHRAN, IRAN

- GPA: -
- Thesis Title: Simulation and Intelligent Control of an Endotracheal Intubation Robot
- Thesis Grade: -
- Supervisor: Dr. Khalil Alipour

Bachelor's Degree, Mechanical Engineering

K. N. Toosi University of Technology

2020–2024

TEHRAN, IRAN

- GPA: 3.79/4 (17.95/20) – 3.81/4 (17.99/20) Last two years
- Thesis Title: Design of a Controller for a Robotic Bicycle Using Intelligent Control Techniques
- Thesis Grade: A (19.00/20)
- Supervisor: Dr. S. Hossein Sadati

High School Diploma, STEM

National Organization for Development of Exceptional Talents (SAMPAD)

2017–2020

RASHT, GUILAN

RESEARCH INTERESTS

- Robotics
- Reinforcement Learning
- Control Theory

TEACHING EXPERIENCE

Teaching Assistant, Machine Vision

Department of Interdisciplinary Sciences and Technologies, University of Tehran

Presented by Dr. Hanieh Naderi

Sep. 2025 – Jan. 2026

- Designed problems for assignments and exams
- Graded the course assignments

TEACHING EXPERIENCE

Teaching Assistant, Automatic Control

Sep. 2023 – Jan. 2024

Mechanical Engineering Department, K. N. Toosi University of Technology

Presented by Dr. Naser Naserifar

- Conducted exercise-solving classes
- Produced educational materials

Teaching Assistant, Computer Programming

Feb. 2022 – Jun. 2022

Mechanical Engineering Department, K. N. Toosi University of Technology

Presented by Dr. Farschad Torabi

- Conducted exercise-solving classes focused on programming in C++ using the Qt framework
- Produced educational materials

TECHNICAL AND RESEARCH EXPERIENCE

- **Physics-informed Data-driven Modeling of a Two-DoF Robot Arm**

Utilized JAX for an optimized implementation of Lagrangian Neural Networks. This model was used for learning the dynamics of a two-DoF robot arm.

- **Simulation and Control of a 5RP Robot Arm** [GitHub]

Design, simulation, and implementation of various control techniques on a 5RP robotic manipulator.

- **Design of a Controller for a Robotic Bicycle Using Intelligent Control Techniques** [GitHub]

Designed linear and nonlinear controllers for a robotic bicycle using the DDPG algorithm. The robot was simulated in MuJoCo.

- **Mobile Robot Control System (teamwork)** [GitHub]

Designed a PID-based control system for a physical differential-drive mobile robot. A GUI application served as a control panel. The user was able to draw the desired path of the robot in the control panel.

- **Passive Steering Wheel (teamwork)** [GitHub]

Designed a CNN model capable of estimating the steering angle from an input image of a steering wheel.

PUBLICATIONS

- **Gray-Box Computed Torque Control for Differential-Drive Mobile Robot Tracking** [arXiv]
- **Optimal Intelligent Control of Robotic Manipulators with Optimal Reliance on Physics Information** [TechRxiv]
- **LiteMish: A Computationally Efficient and Smooth Algebraic Alternative to Mish** [TechRxiv]

TECHNICAL SKILLS

- Programming languages and mathematical packages: C, C++, MATLAB, Python
- Software packages: Git, Proteus, LTSpice, STM32CubeMX, Keil uVision
- Software libraries and frameworks: Keras, TensorFlow, JAX, OpenCV, MuJoCo, Qt (GUI development)
- Computer aided design: SOLIDWORKS
- Development boards: STM32, ESP32, Arduino
- Operating systems: Windows, Linux
- Hardware skills: Soldering
- Other: Deep RL, Simulink, Network Programming, L^AT_EX

COURSES STUDIED

Course Name	Grade (out of 20)
Advanced Robotics	19.83
Machine Vision	20.00
Optimal Control	18.75
Modern Control	18.80
Fundamentals of Electrical Engineering I	20.00
Fundamentals of Electrical Engineering II	19.50
Neural Networks	19.00
Measurement and Control Systems	18.50
Engineering Drawing II	19.60
Introduction to Mechatronics	18.00

CURRENT RESEARCH

Integrating deep reinforcement learning with nonlinear control techniques to design controllers for uncertain robotic systems with stability guarantees

HONORS AND AWARDS

- **Ranked within the top 10% of mechanical engineering students from the same year of entrance**
Issued by K. N. Toosi University of Technology • Sep. 2024
- **Ranked within the top 1.5% in the Iranian University Entrance Exam**
Issued by National Organization of Educational Testing • Sep. 2020

LANGUAGES

- **English**
Proficient
TOEFL iBT Score: 90
- **Persian**
Native language

REFERENCES

Farschad Torabi, Associate Professor

Department of Mechanical Engineering
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Naser Naserifar, Assistant Professor

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S. Hossein Sadati, Associate Professor

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